Advanced JavaScript

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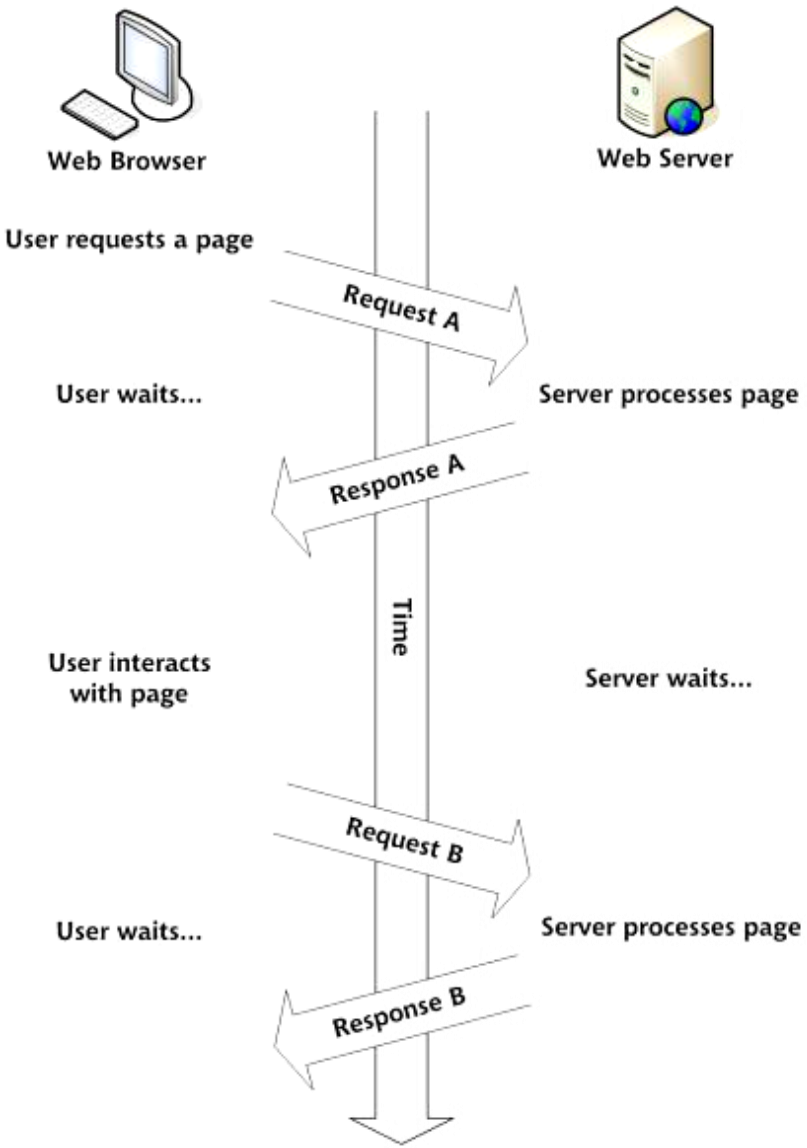
ajax

Ajax

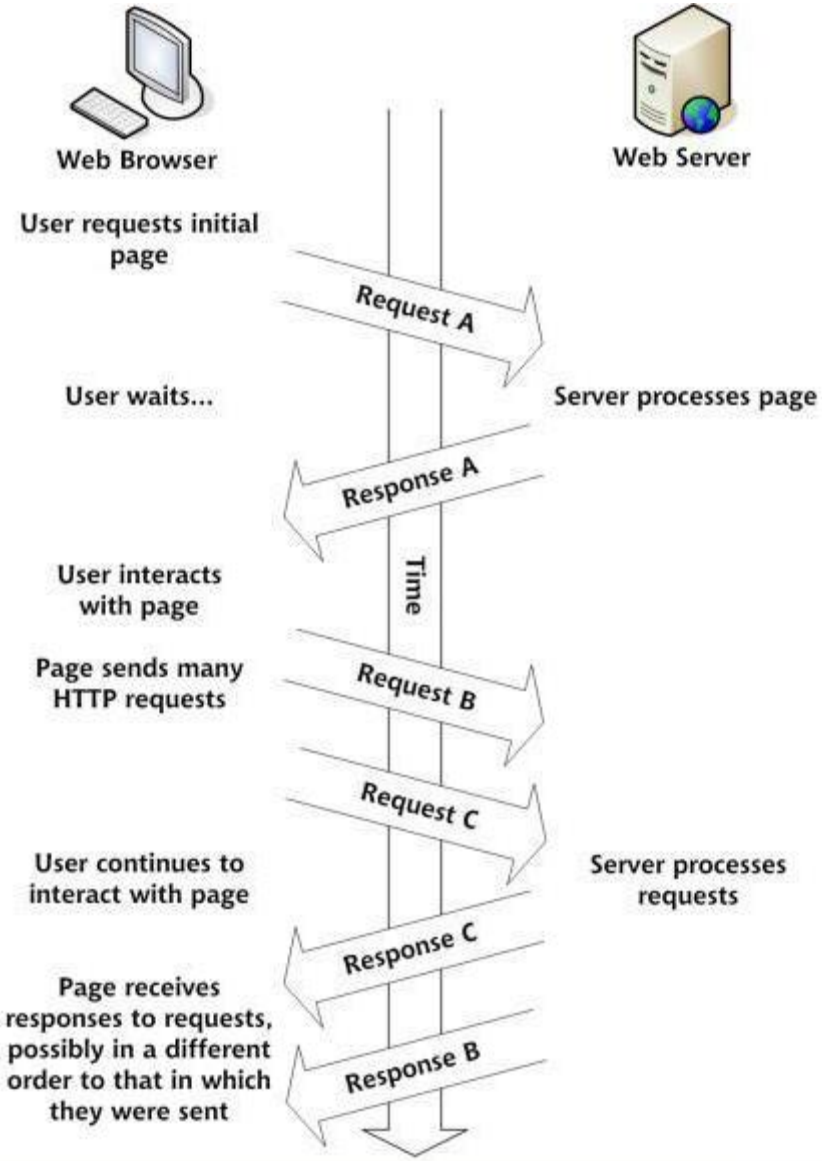
http://adaptivepath.org/ideas/ajax-new-approach-web-applications/

- *Ajax stands for Asynchronous JavaScript and XML.*
- A new paradigm which was introduced in 2005 by Jesse James Garrett.
- The purpose of Ajax is that of giving the illusion that websites are responsive.
- It achieves this by processing requests which involve the sending and receiving of small packets of data without refreshing the web browser.

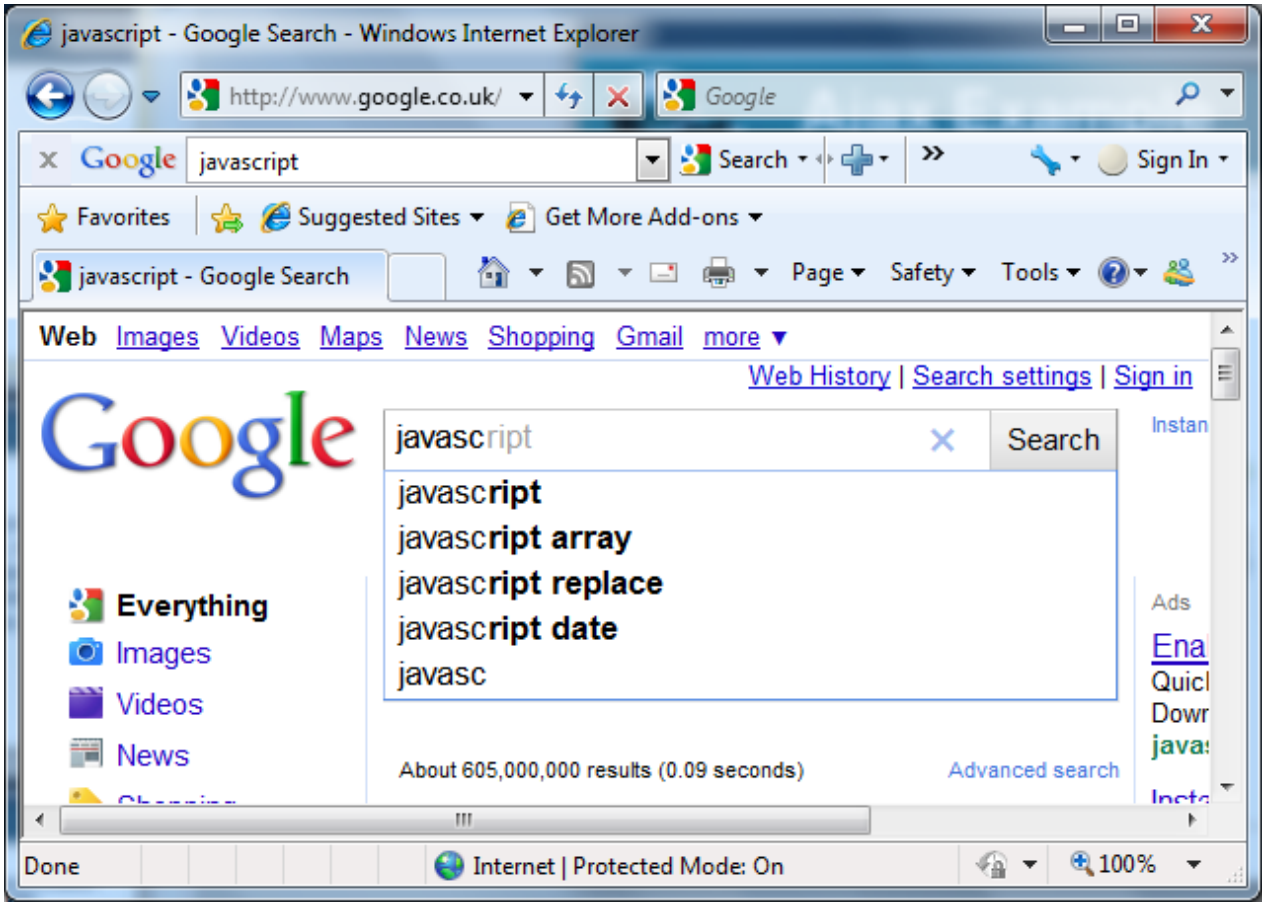
Synchronous



ASynchronous



Ajax Example



Ajax

- Ajax is not a new technology, nor a new programming language.
- It is a methodology for creating fast, rich, user-friendly, and interactive Web applications by allowing a Web page to request small fragments of information from the server instead of an entire page.
- XML is not a requirement for writing Ajax applications

Ajax is founded on

- **JavaScript**
  - With JavaScript, you can make an HTTP request to the server, get the response and update only a part of the page.
- **XHTML**
  - A rigid subset of html which is used to mark-up and style the information.
- **DOM**
  - The Document Object Model which can be accessed by the client browsers.
- **XHR Object**
  - The object used to exchange the information asynchronously.
- **XML**
  - The format used to transfer the data from the server to the client.

XMLHttpRequest Object

XHR

XMLHttpRequest Object

- The XMLHttpRequest object is not specified in any W3C recommendation. (**XHR** for short)
- It is an object (a constructor function) that allows you to send HTTP requests from JavaScript.
- Historically, was introduced in IE and was initially implemented as an **ActiveX** object.
- The XHR object is supported in all modern browsers.

XHR Object

- XHR object across browsers gave birth to the so-called AJAX applications, where it is no longer necessary to refresh the whole page every time you need new content.

- With JavaScript, you can make an HTTP request to the server, get the response and update only a part of the page.

XHR Object Properties/Events

https://developer.mozilla.org/en-US/docs/Web/API/XMLHttpRequest

Properties	Description
onreadystatechange	An event handler for an event that fires at every state change
readyState	Returns the state of the object
responseText	Returns the response as a string
responseXML	Returns the response as XML.
status	Returns the status as a number (e.g. 404 for "Not Found" or 200 for "OK")
statusText	Returns the status as a string (e.g. "Not Found" or "OK")

readyState Values Description

Value	Description
0 XMLHttpRequest.UNSENT	uninitialized (object has been created but not initialized)
1 XMLHttpRequest.OPENED	loading (Loading, send method not yet called)
2 XMLHttpRequest.HEADERS_RECEIVED	loaded (Loaded, send method called, headers not available)
3 XMLHttpRequest.LOADING	interactive (Interactive, some data has been received, status and response headers not available)
4 XMLHttpRequest.DONE	complete (Complete, all data has been received and is available)

Popular Status & Status Text

- 404: Not found → The server found nothing matching the URI given.
- 200: OK → The server successfully returned the page
- 400: Bad Request → Server didn't understand the request due to malformed syntax.
- 401: Unauthorized → The request requires user authentication.
- 500: Internal Server Error → The server encountered an unexpected error and couldn't fulfill the request.
- 503: Service Unavailable → The server is currently unable to handle the request due to temporary overloading or maintenance.

XHR Object Methods

Methods	Description
open("method","URL“ [,async [,"uname“[,"pswd“]]])	Specifies the method, URL, and other optional attributes of a request async default is true
send(content)	Sends the request •***content*** is the posted in a **POST** request •***content*** can be **null** or empty

XHR Object

- The *XMLHttpRequest* object has several important properties and methods that will give you information about
  - the status of the request response,
  - the data that was returned from the server, when the state of the server changes, and so on.
- There are methods to
  - initialize the *XMLHttpRequest* object,
  - send a request,
  - inform you about what is in the response headers,
  - how to cancel a request, and so on.

XHR Object

- There are effectively two steps to using the XMLHttpRequest:
  - Send the request:
    - this includes creating an XMLHttpRequest object and attaching an event listener
  - Process the response:
    - your event listener gets notified that the response has arrived and your code gets busy doing something useful with the response

The Steps for Creating Ajax Communication

- Create an ***XMLHttpRequest*** object.
- Use the object's ***open()*** method to initialize the object
- Send the request to the server with the ***send()*** method.
- Track the state of the request object by ***readyState*** property of ***XMLHttpRequest*** object.

Example!

```
var xhr=new XMLHttpRequest();

xhr.open("GET","fileNM.fileExt");

xhr.onreadystatechange=function(){
    if(xhr.readyState==4)
        if(xhr.status==200){
            var dataObj=JSON.parse(xhr.responseText);
            //start working with recived data
        }
}

xhr.send('');
```

JavaScript Object Notation

JSON

JSON

- JSON stands for JavaScript Object Notation
- Like Ajax, it is not a programming language
- It is a special object notational construct, that is a subset of JavaScript.
- It is not a markup language.
- JSON is based on two data structures:
  - arrays and
  - objects.

JSON

- A Data Interchange Format
- Text-based
- Light-weight
- Simple
- Easy to parse

JSON Structure “Object”

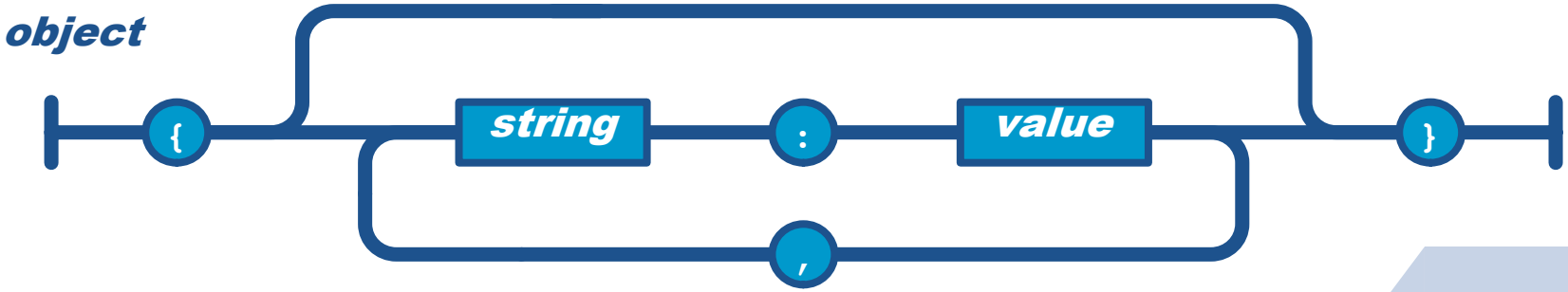
- JSON can be either an Object or an Array.
- The Object is an **unordered** set of name/value pairs.
- An Object begins with ({) and end with (})
  - A pair is a name followed by **(:)** and the value
    - The colon is called the name separator
  - The key/value pairs are separated with a (,)
    - The comma is called value separator.
  - Values are JSON values

JSON Structure “Array”

- JSON Array is an **ordered** sequence of values
- JSON Array begins with ([) and ends with (])
- Values are separated by a comma (,),
- The values can be any of the primitive types as well as the two structures, JSON objects and JSON arrays.

Object

- Objects are unordered containers of key/value pairs
- Objects are wrapped in `{ }`
- `,` separates key/value pairs
- `:` separates keys and values



Object Example

```
{
    "name": “json",
    "at large": true,
    "grade": "A",
    "format": {
"type": "rect",
"width": 1920,
"height": 1080,
"framerate": 24
            }
}
```

```
var personObj={
    “firstName": “json",
    “lastName": true,
    “age": "A",
    “tellYourAge": function {
            alert(“This age is”+this.age);
        }
    }
```

Note: Accessing Data In JSON

Dot Notation→

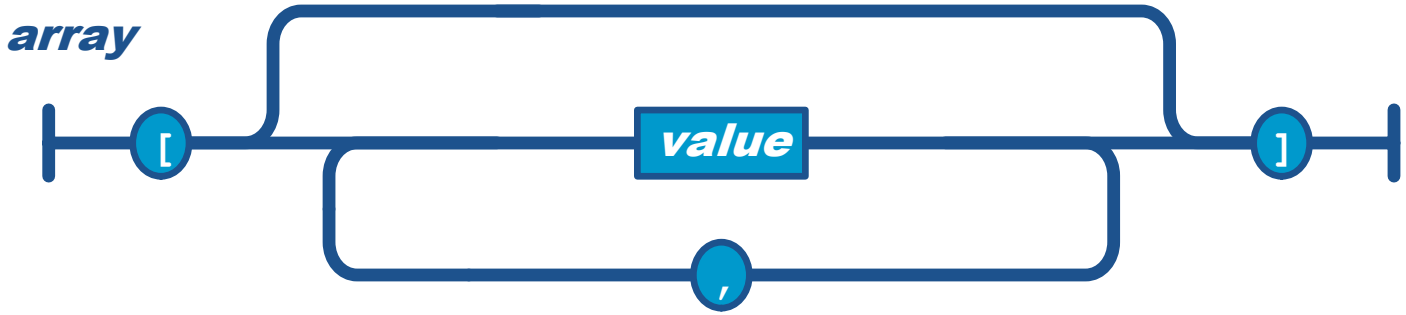
myObj.myField=“something”

Subscript Notation(as Associative Array) →

myObj[‘myField’]=“something”

Array

- Arrays are ordered sequences of values
- Arrays are wrapped in `[]`
- `,` separates values




```
["Sunday", "Monday",
"Tuesday", "Wednesday",
"Thursday", "Friday",
"Saturday"]
```

```
[
    [0, -1, 0],
    [1, 0, 0],
    [0, 0, 1]
]
```

Object Array Example

```
var myJSONObject =
{
    "web":[
        { "name": "html", "years": "5"},
        { "name": "css", "years": "3"}
        ]
    "db":[
          { "name": "sql", "years": "7"}
          ]
}
```

Example!

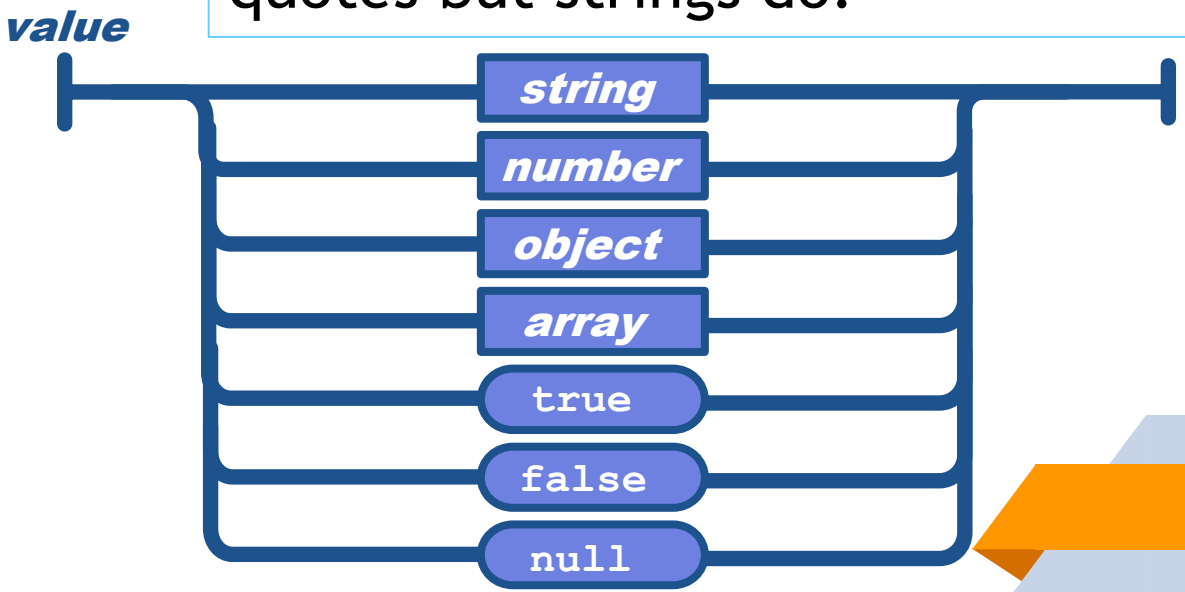
JSON Values

- Strings
- Numbers
- Booleans
  - true
  - false

Unlike JavaScript it does not support hexadecimal and octal representation of numbers, nor does it support NaN (not a number) and infinity.
It supports scientific notation like 6.023e+23. Also numbers do not need quotes but strings do.

- Objects
- Arrays

- `null`



jQuery AJAX Methods

```
$.ajax("std.json",{
    type:"GET",
    dataType:'json',
    contentType:'application/json',
    success:function(dataObj){
        //start working with recieved data
    },
    complete:function(){alert('done')},
    beforeSend:function(){alert('start')},
    //timeout:1,
    error:function(jqxhrObj,errtype,errmsg){
        //handle error occurance
    }

})
```

http://api.jquery.com/category/ajax/

```
var jqxhr = $.getJSON( ".json", function(dataObj) {
    //start working with recieved data
});
```

Assignment